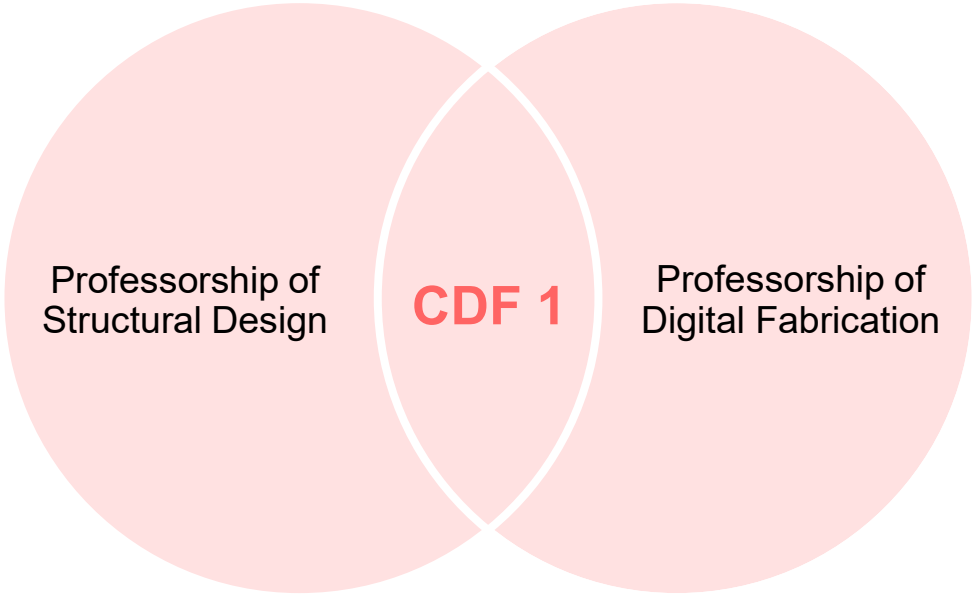
Computational Design and Fabrication 1

Winter Semester 2026/2027 | Mondays 9:45–13:00 | 6 ECTS

Professorship of Structural Design | Professorship of Digital Fabrication

Teaching Team



Teaching Team



Majid Hojjat
TUM Professorship of
Digital Fabrication



Kathrin Dörfler
TUM Professorship of
Digital Fabrication



Pierluigi D'Acunto
TUM Professorship of
Structural Design

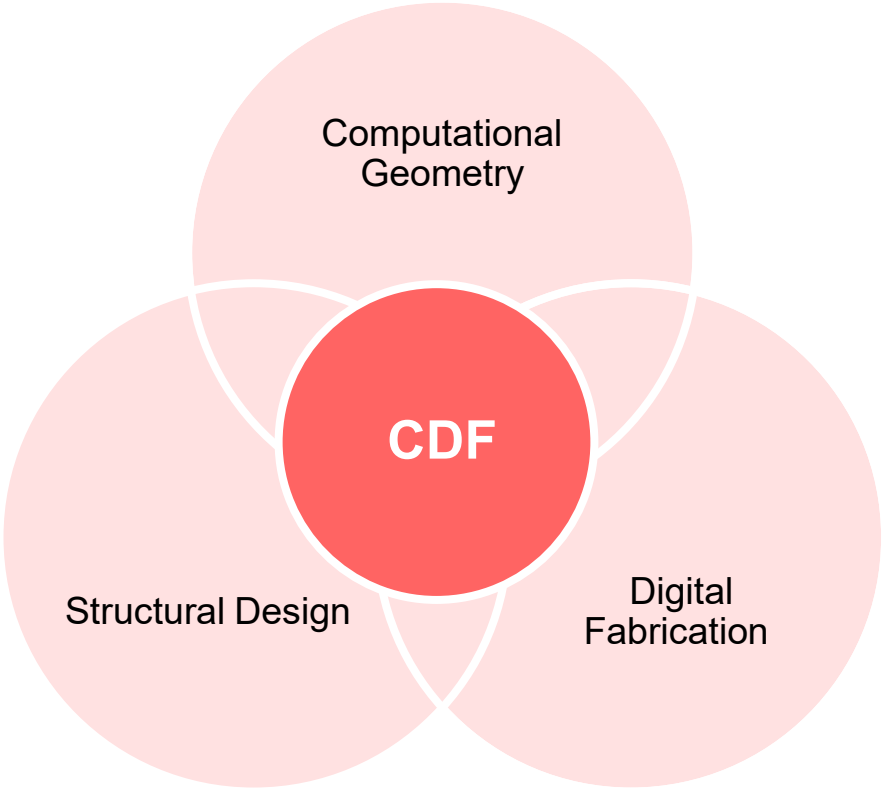


Begüm Saral
TUM Professorship of
Digital Fabrication



Bhavatarini Kumaravel
TUM Professorship of
Structural Design

Motivation

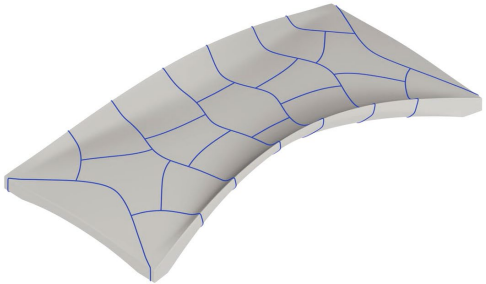

Computational Geometry
CDF
Structural Design
Digital Fabrication


Bridge the gap

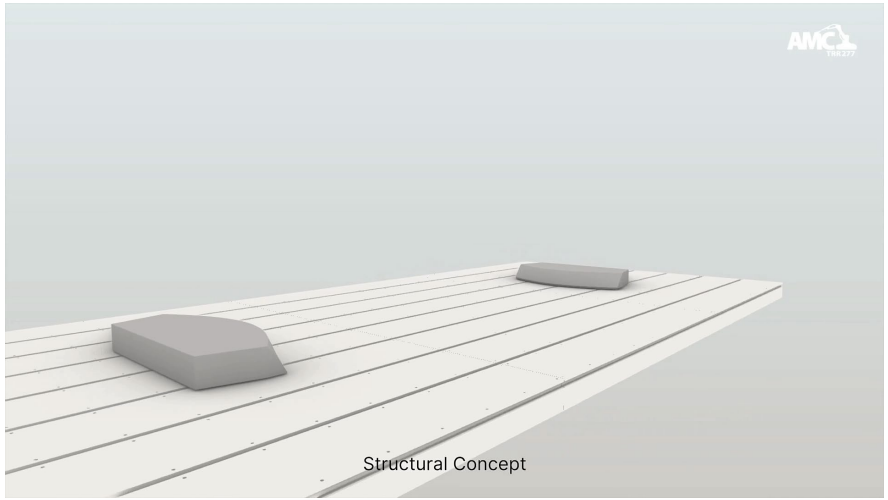

Knitcrete Bridge


Collaborative Robotic Assembly


Intrusion Earth Additive Manufacturing



Segmentation - Principal stress lines
Final segmentation following the principal stress lines and considering a maximum weight of 140kg

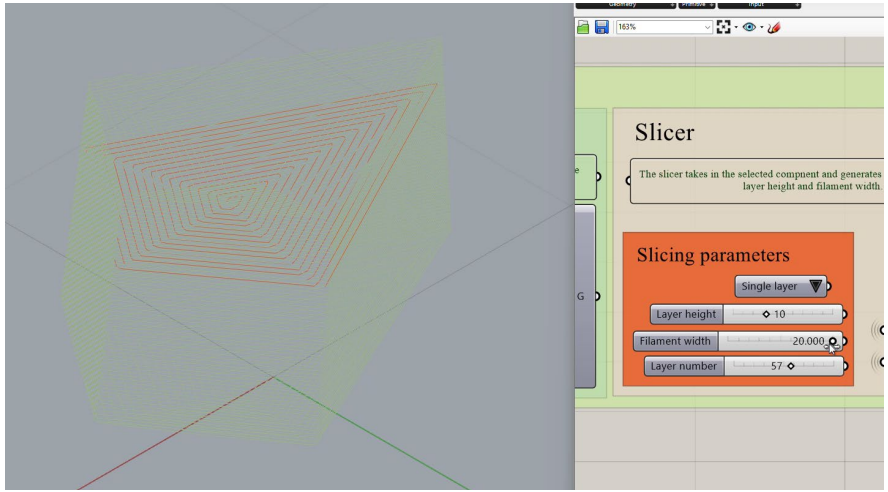


BUILT TOTAL:
3

BUILT BY HUMAN:
3

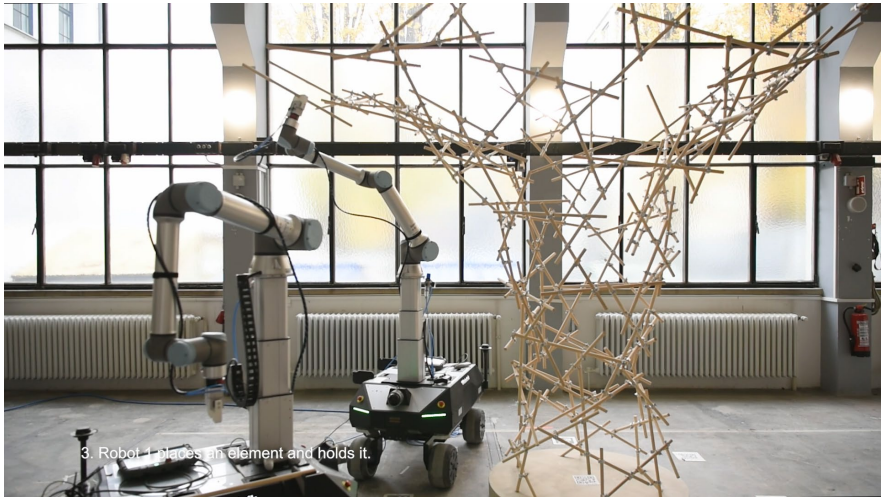
BUILT BY ROBOT:
0






Fabrication by Selective Paste Intrusion at Metallconcept Group / Scawo3d
Printing of the complete bridge during a single day


Spraying thin coat of Cement Paste (Freezing)


3. Robot 1 places an element and holds it.




Bridge the gap


Knitcrete Bridge


Collaborative Robotic Assembly


Intrusion Earth Additive Manufacturing

Learning Goals

1. Understand the basic principles of **computational geometry** and apply them in **parametric and algorithmic design**.
2. Understand the basic principles of **materials and fabrication processes** and develop their **computational representations.**
3. Utilizing **AI-assisted coding** to develop **computational design workflows** for architectural and structural design research and digital fabrication

Course Structure

CDF1 (Part 1)
Parametric & Algorithmic Design

00 Integrated Design & Fabrication Representations

↓

01 Basics of Python

↓

02 Computational Geometry

↓

03 Rule-based design

Course Structure

CDF1 (Part 1)
Parametric & Algorithmic Design

- 00 Integrated Design & Fabrication Representations
- 01 Basics of Python
- 02 Computational Geometry
- 03 Rule-based design

CDF1 (Part 2)
Materials and Processes

- 04 Material and Processes 0: Sand
- 05 Material and Processes 1: Solid Parts
- 06 Material and Processes 2: Clay
- 07 Material and Processes 3: Textile

Course Structure

CDF1 (Part 1)
Parametric & Algorithmic Design

- 00 Integrated Design & Fabrication Representations
- 01 Basics of Python
- 02 Computational Geometry
- 03 Rule-based design

CDF1 (Part 2)
Materials and Processes

- 04 Material and Processes 0: Sand
- 05 Material and Processes 1: Solid Parts
- 06 Material and Processes 2: Clay
- 07 Material and Processes 3: Textile

CDF2
(optional)

- Introduction: Semester topic
- Hands-on Experimentation
- Design Development
- 1:1 Fabrication

Course Schedule: Part 1 – Parametric & Algorithmic design

	Date	Topic	Content			Assignment
1	12 October 2026	**Introduction + Registration**	Lecture 0.0 Course Introduction	Registration		
2	19 October 2026	**Integrated Representations + Getting Started with Python**	Lecture 1.0 Introduction to Integrated Representations	Lecture 1.1 Basics of Python	Hands-on: Simple Python Copilot-based vibe coding	
3	26 October 2026	**Computational Geometry**	Lecture 2.0 Introduction to Computational Geometry	Hands-on: Copilot-based vibe coding for geometry generation		**Assignment 1**
4	2 November 2026		Feedback session			
5	9 November 2026	**Rule-based design**	Lecture 3.0 Rule-based design	Hands-on (Manual): Patterns with algorithmic logic	Hands-on (Coding): Copilot-based vibe coding	**Assignment 2**
6	16 November 2026		Feedback session			

Course Schedule: Part 2 – Materials & Processes

	Date	Topic	Content			Assignment
7	23 November 2026	**Material and Processes 0: Sand (Modelling, Simulation, Representation)**	Lecture 4.0 Material and Processes 0: Sand	Hands-on (Manual): Making with materials	Hands-on (Coding): Groups working on digital representations	**Assignment 3**
8	30 November 2026		Feedback session			
9	7 December 2026	**Material and Processes 1: Solid Parts - Modular Component Assembly (Tectonics)**	Lecture 5.0 Material and Processes 1: Solid parts	Hands-on (Manual): Making with materials	Hands-on (Coding): Groups working on digital representations	**Assignment 4**
10	14 December 2026		Feedback session			
11	11 January 2027	**Material and Processes 2: Clay - Freeform**	Lecture 6.0 Material and Processes 2: Clay	Hands-on (Manual): Making with materials	Hands-on (Coding): Groups working on digital representations	**Assignment 5**
12	18 January 2027		Feedback session			
13	25 January 2027	**Material and Processes 3: Textile**	Lecture 7.0 Material and Processes 3: Textile	Hands-on (Manual): Making with materials	Hands-on (Coding): Groups working on digital representations	**Assignment 6**
14	1 February 2027		Feedback session			

Evaluation

Assignment	Title	Evaluation (%)
Assignment 1	Computational Geometry	10%
Assignment 2	Rule-based Design	10%
Assignment 3	Materials & Processes 0: Sand	20%
Assignment 4	Materials & Processes 1: Solid parts	20%
Assignment 5	Materials & Processes 2: Clay	20%
Assignment 6	Materials & Processes 3: Textile	20%

Attendance

Attendance is mandatory. Students may miss up to two sessions (15% of the classes) in total.
However, only one of these may be an input (lecture) session.
Missing more than this will result in not passing the course.

Attendance Form for today’s Lecture

Scan the QR code or open the link below to register your attendance.

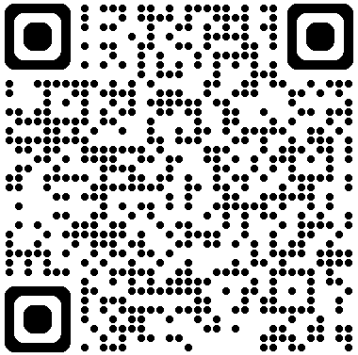
https://forms.cloud.microsoft/e/6GU0WiUsu2

Use your TUM ID to sign in and access the form.

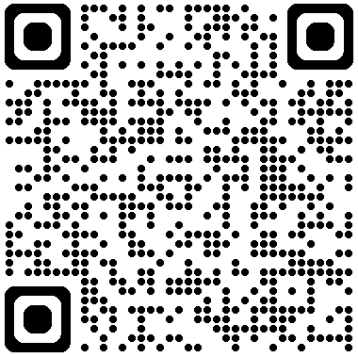


Teaching Materials

Moodle



Github




We look forward to the semester CDF 2026/27!